



SOAR & SIEM Tutorial: A Complete Guide

A comprehensive guide to SIEM (Security Information and Event Management) and SOAR (Security Orchestration, Automation, and Response) — the backbone of modern Security Operations Centers. From log collection and correlation rules to playbook automation and incident response, this book explains concepts, architecture, real-world workflows, and hands-on exercises for building a career in security operations.

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1. Security Operations & the SOC

CHAPTER 1

What is Security Operations?

Security operations is the set of processes, people, and technologies an organization uses to detect, investigate, respond to, and recover from cyber threats. It is the operational side of cybersecurity — the "always-on" layer that watches the environment, reacts to suspicious activity, and reduces the impact of attacks. The place where security operations happen is called a **Security Operations Center (SOC)**.

A SOC is not just a room full of monitors. It is a coordinated team of analysts, engineers, and managers supported by a technology stack — of which **SIEM** (for visibility) and **SOAR** (for action) are the two most important components. Understanding how a SOC works is the essential foundation for everything else in this book.

The Security Operations Lifecycle

Security operations follow a continuous cycle. Different frameworks describe it slightly differently, but the core loop is the same:

Phase	What Happens	Primary Tool
Prepare	Define policies, procedures, use cases, and playbooks; staff and train the team	Documentation, SOAR playbooks
Collect	Gather logs and telemetry from across the environment	SIEM, log management, agents
Detect	Correlate data to identify suspicious activity and generate alerts	SIEM correlation rules, detection engineering

Triage	Prioritize alerts, filter noise, determine severity and impact	SOAR, analyst queues, enrichment
Investigate	Dig into the alert, gather evidence, scope the incident	SIEM search, threat intel, case management
Respond	Contain, eradicate, and recover; block IOCs, isolate hosts, restore systems	SOAR automation, EDR, firewalls
Learn	Post-incident review, tune detections, update playbooks	Metrics, reporting, continuous improvement

Key Idea: The SOC lifecycle is a loop, not a line. Every incident teaches the team something that improves preparation, detection, and response for the next one. Mature SOC's run this loop continuously and measure every stage.

2. SIEM Fundamentals

CHAPTER 2

What is a SIEM?

SIEM stands for **Security Information and Event Management**. It combines two older disciplines into one platform:

- **SIM (Security Information Management):** long-term storage, analysis, and reporting of security data — the "information" half.
- **SEM (Security Event Management):** real-time monitoring, correlation, and alerting on security events — the "event" half.

A SIEM centralizes logs and telemetry from across the environment into one place, normalizes them into a common format, correlates them to spot suspicious activity, alerts analysts when something looks wrong, and provides search, dashboards, and reporting for investigation and compliance.

Core Components of a SIEM

Every SIEM, from open source to enterprise, shares the same conceptual pipeline:

Component	Function
Collector / Forwarder	Gathers logs and events from source systems (servers, firewalls, cloud, endpoints) and sends them to the SIEM
Ingestion / Pipeline	Receives data, parses raw formats, validates, and routes records to storage
Parser / Normalizer	Translates many different log formats into a single normalized schema so events can be compared

Storage / Index	Persists normalized events in a fast-searchable format (inverted index, data lake, or both)
Correlation Engine	Applies rules to detect combinations and sequences of events that indicate an attack
Alerting Engine	Generates alerts when correlation rules fire; routes them to queues, email, or SOAR
Search & Dashboards	Lets analysts query the data and build visualizations for monitoring and investigation
Reporting	Produces compliance and management reports (e.g., PCI DSS, ISO 27001, internal metrics)

Key Idea: A SIEM is fundamentally a *centralized log analytics platform built for security*. The value comes not from storing logs, but from normalizing them into a common schema so a single correlation rule can see an attack that spans many different systems.

3. SIEM Architecture & Data Pipeline

CHAPTER 3

Anatomy of a SIEM Data Pipeline

The heart of any SIEM is its data pipeline: the series of stages that move raw logs from source systems into normalized, searchable, actionable events. Understanding each stage matters because most SIEM problems — missing data, slow searches, bad correlations, alert gaps — are actually pipeline problems.

Stage	Description	Example Technologies
1. Log generation	Applications, OSes, network devices produce raw log records	syslog, Windows Event Log, JSON APIs, cloud audit logs
2. Collection	Agents or forwarders pull or push logs to the SIEM	Winlogbeat, Filebeat, Splunk Universal Forwarder, Azure Agent, Syslog-ng
3. Transport	Logs travel over the network, often with encryption and buffering	TLS, TCP/UDP, AMQP, Kafka, HTTP ingestion APIs
4. Parse & normalize	Raw text/JSON is parsed into structured fields on a common schema	CEF, LEEF, ECS, KV format, Grok, regex rules
5. Enrich	Add context: geo-location, asset owner, user info, threat-intel tags	GeoIP, CMDB/asset DB, Active Directory, TI feeds

6. Index / store	Normalized events are indexed and stored for fast search	Lucene/inverted index, columnar store, hot/warm/cold tiers
7. Correlate & alert	Rules process events in real time and emit alerts	Correlation engine, detection rules, alert rules
8. Investigate	Analysts search, visualize, and pivot through the data	Query language, dashboards, notebooks

Common Log Sources

A SIEM is only as good as the data you feed it. The most important sources, roughly in order of security value:

- **Identity & authentication:** Active Directory/Entra ID, Okta, VPN, MFA — detect credential abuse, account lockouts, privilege changes.
- **Endpoints:** Windows Event Logs (Security, Sysmon), EDR telemetry, Linux auditd — the primary host-level visibility.
- **Network:** Firewalls, proxies, DNS, NetFlow — detect scanning, C2 beaconing, data exfiltration.
- **Servers & applications:** Web servers (IIS, nginx), databases, application logs — detect web attacks and data access.
- **Cloud:** AWS CloudTrail, Azure Activity Log, Google Cloud Audit Logs — cloud-native events, API abuse, IAM changes.
- **Email:** Exchange/M365 message tracking, email gateway logs — phishing detection.
- **Antivirus / EDR / EPP:** Malware detections, blocked threats — high-fidelity signals.

Design Principle: Map log sources to the *use cases* you want to detect before building ingestion. If you cannot articulate what attack you will detect with a log source, question why you are collecting it. This keeps cost and noise under control.

4. SIEM Features & Capabilities

CHAPTER 4

Search: The Foundation Skill

Everything else in a SIEM builds on search. Analysts spend much of their day querying data to confirm or refute hypotheses. Each platform has its own query language — but they share concepts. The most common are:

Language	Platform	Example
SPL (Search Processing Language)	Splunk	<code>index=wineventlog EventCode=4625 stats count by User</code>
KQL (Kusto Query Language)	Microsoft Sentinel / Azure	<code>SecurityEvent where EventID == 4625 summarize count() by Account</code>
Lucene	Elastic	<code>event.code:4625 AND source.ip:10.0.0.0/8</code>
Ariel / AQL	IBM QRadar	<code>SELECT COUNT(*) FROM events WHERE eventID=4625</code>
SQL-like	Chronicle, others	<code>SELECT * FROM events WHERE ...</code>

```
# KQL example: all failed logons from external IPs in the last hour
SecurityEvent
| where TimeGenerated > ago(1h)
| where EventID == 4625
| where IPAddress !startswith "10." and IPAddress !startswith "192.168."
| project TimeGenerated, Account, IPAddress, WorkstationName
| sort by TimeGenerated desc
```

Good query skills include: knowing the schema, using time ranges correctly, pivoting between entity fields (user, host, IP), and avoiding full-table scans where indexes exist.

Correlation Rules

A correlation rule is the core detection unit: a query or event pattern that, when matched, produces an alert. Rules vary from simple single-event filters to complex multi-stage, stateful correlations that track a sequence over time.

A well-formed correlation rule has several parts:

- **Trigger conditions:** which events, fields, and thresholds.
- **Time window:** how long events must occur within (e.g., 10 failed logins in 5 minutes).
- **Grouping:** key events are correlated by (e.g., same user, same source IP).
- **Severity mapping:** informational → critical based on impact.
- **Suppression:** avoid alert storms (e.g., one alert per 15 minutes per host).
- **Response action:** who is notified, what automation runs, which SOAR playbook fires.

Example rule (concept): "*Brute Force Attempt*" — fire when the same user has 10+ failed logins (`EventID 4625`) from the same IP within 5 minutes, severity Medium, suppress for 15 minutes, escalate to SOAR for IP enrichment.

5. Detection Engineering for SIEM

CHAPTER 5

What is Detection Engineering?

Detection engineering is the discipline of designing, implementing, testing, and maintaining the detections that turn raw security data into meaningful alerts. It is the bridge between "we have a SIEM" and "we can reliably detect attacks." A detection engineer decides what to detect, writes the rule, tests it, tunes it, and decommissions it when it no longer earns its place.

From Threat Model to Use Case

Good detection work starts before any rule is written. The process:

1. **Know your threats:** map the organization's crown-jewel assets and the realistic attack paths to them (MITRE ATT&CK is the de facto framework).
2. **Define a use case:** a plain-language statement of what you want to detect and why. Example: "Detect credential dumping via LSASS access on Windows endpoints."
3. **Identify data:** which log sources and fields provide visibility into this behavior.
4. **Write the detection:** encode the use case as a rule, query, or analytics model.
5. **Test and validate:** prove it fires on known-bad data and stays quiet on normal traffic.
6. **Deploy and tune:** ship, monitor false positives, iterate.
7. **Retire:** review rules periodically; delete what is ineffective.

MITRE ATT&CK as a Detection Backbone

MITRE ATT&CK is a free, globally accessible knowledge base of adversary tactics and techniques. Tactics are the "why" (Initial Access, Execution, Persistence, Privilege Escalation, Defense Evasion, Credential Access, Discovery, Lateral Movement, Collection, C&C, Exfiltration, Impact); techniques are the "how" (e.g., T1003.001 OS Credential Dumping: LSASS Memory).

Detection engineers use ATT&CK to:

- Prioritize techniques that map to the organization's risk.
- Reference documented behaviors, examples, and detections from the community.
- Track coverage: which techniques you can detect, and which are gaps.
- Design "detect the behavior, not the signature" — technique-focused rules survive malware family churn.

Analogy: Rules tied to a specific malware sample are like a wanted poster for one person. Rules tied to a technique (e.g., "LSASS access without a legitimate reason") are like a description of the behavior that catches anyone doing it — even new criminals.

6. SOAR Fundamentals

CHAPTER 6

What is SOAR?

SOAR stands for **Security Orchestration, Automation, and Response**. Coined by Gartner, it describes platforms that help security teams coordinate tools and processes, automate repetitive tasks, and execute incident response faster and more consistently.

The three words describe three distinct capabilities that a SOAR platform combines:

Capability	What it does	Example
Orchestration	Connects and coordinates multiple security tools via integrations and APIs	SIEM hands an alert to a SOAR, which queries an EDR and a threat-intel feed
Automation	Executes routine, predefined actions without human intervention	Automatically quarantine a file hash on every endpoint
Response	Manages the end-to-end incident workflow: cases, tasks, evidence, approvals, documentation	Create a case, assign it to an analyst, track MTTR, close with a full timeline

Key Idea: A SOAR is not just "automation." Automation without orchestration is a script that can't reach the tools. Orchestration without automation is a fancy integration framework. Response without either is a ticketing system. The value is the combination, plus the governance and audit trail that make it trustworthy.

7. SOAR Platforms & Architecture

CHAPTER 7

Anatomy of a SOAR Platform

Although SOAR vendors differ in UX and pricing, they share a common architecture. Understanding these layers lets you reason about any platform and design playbooks effectively:

Layer	Components	Purpose
Ingestion	Alert connectors, API/webhook listeners, email listeners, SIEM integrations	Bring alerts and incidents into the platform as data objects
Orchestration	Connector library, action definitions, API gateway, credential vault	Call external tools securely and uniformly
Playbook engine	Visual workflow builder, triggers, actions, conditions, loops, error handling	Define and execute automation workflows
Case management	Incidents, tasks, evidence, timelines, assignments, statuses	Manage the human side of response
Integration & data	Threat-intel stores, asset/CMDB links, artifact normalization	Provide context (IOCs, users, assets) to playbooks
Governance & access	RBAC, audit logs, approvals, separation of duties	Ensure only authorized automation and people act
Analytics & metrics	Reports, dashboards, MTTR/MTTA tracking, automation rate	Prove value and drive tuning

Key Concept — Artifacts: In SOAR platforms, an *artifact* is a discrete piece of evidence extracted from an alert — an IP, domain, file hash, email address, or username. Playbooks act on artifacts. Normalizing artifacts (e.g., always extracting `file.sha256`, `destination.ip`) is what makes enrichment and automation reusable.

8. Playbook Development

CHAPTER 8

What is a Playbook?

A **playbook** is an executable, machine-readable workflow that defines how the SOAR responds to a specific trigger — usually an alert type or an incident condition. It is the automation of a documented incident-response procedure: the difference is that the SOAR can execute it deterministically, log every step, and handle the routine parts without a human.

Playbook vs Runbook: A *runbook* is human documentation — "if X happens, do steps A, B, C." A *playbook* is that procedure encoded into automation. Most mature SOC's start with runbooks, then convert the highest-volume, most-repetitive ones into playbooks.

9. Security Automation in Action

CHAPTER 9

Where Automation Delivers the Most Value

Not everything should be automated. The best candidates share these traits: high volume, well-defined decision logic, repetitive steps, clear data, low ambiguity, and (ideally) low blast radius. The canonical winners are alert triage, phishing response, malware analysis, IOC blocking, and deduplication.

1. Automated Alert Triage

Alert triage is the biggest time sink in most SOCs, so it is the first automation target. A triage playbook typically:

1. Receives the alert from the SIEM.
2. Normalizes artifacts and looks up the affected asset's criticality.
3. Queries threat intelligence for each IOC.
4. Scores the alert (e.g., 0-100) based on reputation, criticality, and behavior.
5. Auto-closes clearly benign alerts with a documented reason.
6. Escalates suspicious alerts to a case with all context prepopulated.

Result: Analysts only see the top-scored alerts, with enrichment already done. Typical reductions: 30-70% of alerts never reach a human, and the rest are resolved faster because context is gathered automatically.

10. Integrating SIEM & SOAR

CHAPTER 10

The SIEM + SOAR Partnership

SIEM and SOAR are designed to work as a team. The SIEM is the **eyes**: it sees everything, normalizes it, and detects anomalies. The SOAR is the **hands**: it takes action, orchestrates other tools, and manages the response. The integration between them is where most of the operational value lives.

The Flow: SIEM collects → SIEM detects → SOAR orchestrates → SOAR responds → results feed back to SIEM for validation and tuning. This creates a closed loop: detection informs action, and action informs detection.

11. Threat Intelligence Integration

CHAPTER 11

What is Threat Intelligence?

Threat intelligence (TI) is evidence-based knowledge about adversaries — their capabilities, infrastructure, and methods — that helps an organization detect, prioritize, and respond to threats. In practice, most SOCs use two kinds:

- **Strategic intelligence:** high-level trends, adversary profiles, industry risk (for leadership and planning).
- **Tactical / operational intelligence:** specific indicators of compromise (IOCs), tools, techniques, and procedures (TTPs) that feed detections and playbooks.

For SIEM and SOAR, the tactical layer matters most: it is what gets queried during enrichment and matched against events.

Indicators of Compromise (IOCs)

IOCs are the concrete data points used to detect compromise:

IOC Type	Example	How It's Used
IP address	203.0.113.45	Block on firewall, match in SIEM connection logs
Domain / hostname	evil-domain.example	Block on DNS/proxy, match in email logs
URL	https://evil-domain.example/rz	Block on web filter, scan in phishing

File hash (MD5/SHA1/SHA256)	e3b0c442... (SHA256)	Reputation check, block in EDR/AV
Email address / sender	bad@evil.example	Quarantine in email gateway
Mutual TLS certificate	SHA256 fingerprint	Detect C2 infrastructure reuse

Important nuance: An IP can change hands or be legitimate infrastructure shared with benign users. Hashes expire quickly as malware changes. Domains are dynamic (DGA, fast-flux). Good intelligence programs treat every IOC as a *temporal, confidence-weighted* fact, not an eternal truth.

12. Cloud, XDR & the Modern SOC

CHAPTER 12

The Changing SOC Landscape

Security operations used to mean a data center, a firewall, and a room of analysts. Today's SOCs defend hybrid and multi-cloud environments: SaaS applications, containers, serverless, IoT, and remote workers. This changes what SIEM and SOAR must do:

- Data is scattered across cloud providers and SaaS vendors — each with its own APIs and audit logs.
- Attackers abuse cloud identities, exposed storage, and misconfigured services as often as traditional malware.
- Speed matters more: cloud attacks move at machine speed, so response must too.

XDR: Extended Detection and Response

XDR (Extended Detection and Response) collects and correlates telemetry across the entire attack surface — endpoints, networks, cloud workloads, email, and identities — and automates response. Vendors position XDR as either a complement to or a replacement for parts of the SIEM/SOAR stack.

Dimension	SIEM	XDR	SOAR
Primary role	Centralized log analytics & detection	Cross-domain detection & automated response	Orchestration, automation, case management
Data focus	All ingested logs	Telemetry from connected products (often its own agents)	Alerts and incidents from upstream tools

Strength	Breadth, compliance, custom analytics	Deep endpoint/network correlation, built-in response	Workflow automation across any tool
Vendor lock-in	Medium	Higher (best in same-vendor stacks)	Low (integrates everything)

A common modern architecture: XDR does real-time endpoint/cloud detection and response; SIEM provides the central log platform, compliance, and long-term analytics; SOAR orchestrates across both and beyond. The lines blur — Microsoft Defender XDR + Sentinel, and CrowdStrike Falcon + cloud SIEM partnerships, are examples.

Cloud-Native SIEM: Microsoft Sentinel as a Reference

Sentinel is the archetype of the cloud-native SIEM. Its design illustrates the modern approach:

- **On-demand scale:** pay per GB of ingested data; scale without hardware planning.
- **KQL analytics:** detection rules, hunting queries, and notebooks over the same data.
- **Built-in automation:** playbooks using Azure Logic Apps; automation rules route, tag, and respond.
- **Threat intelligence:** native STIX/TAXII ingestion into watchlists.
- **UEBA & AI:** entity behavior analytics and ML-based alert fusion that groups alerts into incidents.
- **M365/Azure-native coverage:** unmatched visibility into Entra ID, Defender, and Azure activity.

Takeaway: Whether you choose Sentinel, Chronicle, Splunk Cloud, or another platform, the cloud-native model means: consumption pricing, elastic scale, API-first design, and automation built in. If your SIEM can't be driven by API from your SOAR and CI pipelines, it is an old design.

13. Practice, Labs & Career Path

CHAPTER 13

Learn by Building

Security operations is a hands-on discipline. Reading about correlation rules or playbooks is not the same as writing them, breaking them, and fixing them. This chapter gives you a realistic path to practical skill using free or cheap tools — and a roadmap for turning that skill into a career.

Building a Home SOC Lab

A capable learning environment can be built almost entirely with free tools:

Layer	Free / Low-Cost Options	Role
Log sources	Your own Windows/Linux machines, a small VM or Raspberry Pi, a home router	Generate real telemetry
Endpoint telemetry	Sysmon (Windows), auditd (Linux), osquery	Rich host events
SIEM	Elastic Stack (free), Wazuh, Splunk Free	Collect, normalize, correlate, alert
SOAR	Shuffle, TheHive + Cortex	Playbooks, enrichment, case management
Threat intel	AlienVault OTX, AbuseIPDB, VirusTotal Community	IOC enrichment

Attack simulation	Atomic Red Team, known test hashes (eicar)	Safely trigger detections
VMs	VirtualBox / Hyper-V / WSL	Isolated test environment

Lab Tip: A realistic mini-lab: one Linux VM running Elastic or Wazuh as the SIEM; your host (or a Windows VM) shipping Sysmon logs to it; Shuffle as the SOAR receiving SIEM alerts and calling a free enrichment API; Atomic Red Team to generate a few attack events; then build a playbook that triages one alert end to end.

14. References & Resources

CHAPTER 14

SOAR & SIEM Cheatsheet

Concept	Quick Reference
SIEM	Security Information and Event Management — centralized log analytics, correlation, alerting
SOAR	Security Orchestration, Automation, and Response — connects tools, automates tasks, manages cases
SOC	Security Operations Center — the team/process layer that runs detection and response
MTTD	Mean Time to Detect — attack start to detection
MTTA	Mean Time to Acknowledge — alert to analyst engagement
MTTR	Mean Time to Respond/Resolve — detection to containment/resolution
EPS	Events Per Second — the standard SIEM throughput unit
Use case	A plain-language statement of what you want to detect and why
Correlation rule	A detection pattern that produces an alert when matched
Sigma	Portable, open-source detection rule format convertible to many SIEM languages

Detection as code	Managing detection rules in git with CI testing and version control
Playbook	Executable automation of an incident-response procedure
Runbook	Human-readable incident-response documentation
Artifact	A discrete piece of evidence from an alert (IP, hash, domain, user)
Connector	A SOAR integration exposing actions to call an external tool
IOC	Indicator of Compromise — IP, domain, hash, etc. tied to malicious activity
STIX	Structured Threat Information Expression — JSON format for threat data
TAXII	Transport protocol for exchanging STIX intelligence
UEBA	User and Entity Behavior Analytics — ML-based baseline deviation detection
XDR	Extended Detection and Response — cross-domain detection and automated response
MITRE ATT&CK	Adversary tactics and techniques knowledge base
FP / TP	False Positive (benign, matched) / True Positive (real malicious)
Automation rate	% of alerts handled without human intervention
Blast radius	The potential damage if an automated action fails or misfires

Common SIEM Query Language Patterns

Pattern	Splunk (SPL)	Sentinel (KQL)	Elastic (Lucene)
Filter events	<code>index=main</code> <code>sourcetype=wineventlog</code>	<code>Event where ...</code>	<code>source:winlogbeat*</code>
Count by field	<code> stats count by User</code>	<code> summarize count() by Account</code>	<code> top User</code> (or agg)
Time filter	<code>earliest=-1h</code>	<code>where TimeGenerated > ago(1h)</code>	<code>@timestamp:[now-1h TO now]</code>
Sort results	<code> sort - _time</code>	<code> sort by TimeGenerated desc</code>	default score/time sort

Glossary

Alert	A notification generated when a detection condition matches
Case / Incident	A managed record of an investigation with evidence, tasks, and status
CEF	Common Event Format — key=value event encoding used across vendors
CSPM	Cloud Security Posture Management — detects cloud misconfigurations
Enrichment	Adding context (reputation, geo, asset info) to an alert or artifact
False Positive	An alert that fired but represents benign activity
Ingestion	The process of receiving, parsing, and storing event data
Normalization	Mapping diverse log formats to a common schema
Playbook	Executable automation of an incident-response procedure

Severity	A rating of an alert's importance (informational → critical)
Threat Hunting	Proactive search for threats that automated rules missed
Watchlist	A maintained list used to enrich or prioritize events
Zero Trust	"Never trust, always verify" security model

Useful Resources

- **Frameworks:** MITRE ATT&CK (attack.mitre.org), NIST SP 800-61 Incident Handling Guide, NIST Cybersecurity Framework.
- **Detection:** [SigmaHQ/sigma](https://sigma.hq), Atomic Red Team, MITRE ATT&CK Navigator, Sigma conversion tools (pySigma, uncoder.io).
- **Learning platforms:** TryHackMe, LetsDefend, CyberDefenders, Blue Team Labs Online, DetectionLab.
- **Open-source tools:** Elastic Security, Wazuh, Shuffle, TheHive + Cortex, dfir-iris, MISPL, OpenCTI.
- **Vendor docs:** Microsoft Sentinel (Ninja training, free tier), Splunk e-learning, XSOAR docs, Palo Alto Cortex documentation.
- **Communities:** Detection Engineering community, r/securityoperations, ISSA/OWASP chapters, DEF CON Blue Team Village.

Key Industry Reports & Reading

- Gartner Market Guides for SIEM and SOAR (updated annually).
- Verizon DBIR (Data Breach Investigations Report) — real-world attack patterns.
- Microsoft Digital Defense Report — threat landscape and trends.
- MITRE ATT&CK evaluations — detection capability benchmarks.
- SANS SOC survey reports — team structure and tooling benchmarks.

Congratulations! You have completed the SOAR & SIEM Tutorial. You now understand how the SOC operates, how a SIEM turns raw logs into detections, how a SOAR turns

detections into orchestrated response, and how they integrate to shrink MTTD, MTTA, and MTTR. Keep building your lab, keep writing detections and playbooks, and keep measuring — the field rewards hands-on practitioners.